

HealthConnect Clinic Experience Lab

Exploratory Analysis, KPI Development & Business Insights

Week 5 Business Analytics Report

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Executive Summary

This Week 5 analysis examined 5,000 fictional HealthConnect appointment records to understand patterns associated with attendance and no-shows. Of the 4,737 appointments classified as either Attended or No-Show, 51.2% resulted in a no-show. The strongest observed patterns were linked to previous no-show history, booking lead time, and distance to the clinic. Reminder status also showed a meaningful difference, while appointment time, age group, and waiting time showed smaller or less stable differences.

51.2% Overall no-show rate	57.8% Prior no-show history	63.9% 31-60 day lead time	60.5% 20+ km distance
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The analysis supports a targeted operational response rather than a single broad intervention. HealthConnect could test additional confirmation for patients with previous no-shows, reconfirm long-lead bookings closer to the appointment date, review reminder timing and channel strategy, and consider administrative support for patients travelling longer distances.

Business Context and Objective

HealthConnect Clinic is an appointment-based fictional healthcare provider seeking to reduce missed appointments, improve appointment-slot utilisation and strengthen administrative support. Week 4 established the initial problem, reviewed the dataset and data dictionary, identified relevant variables, proposed business questions, and defined potential KPIs. Week 5 moved that foundation into practical analysis.

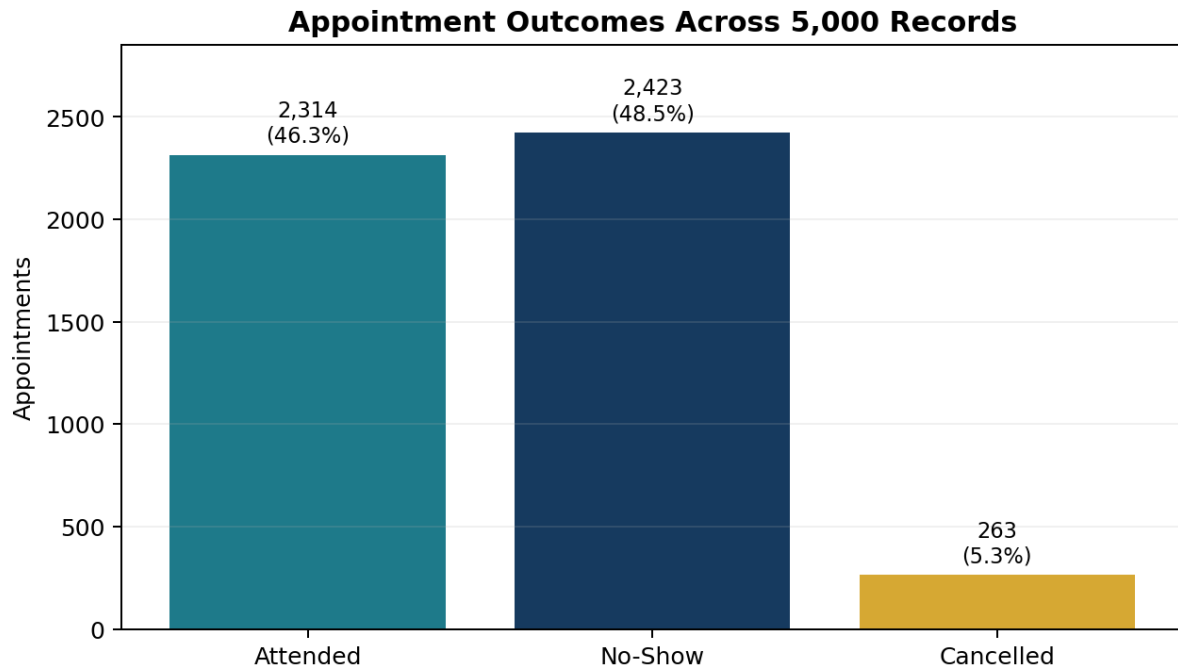
Central analytical question

What patterns in the HealthConnect appointment data are associated with no-shows, and which areas should the clinic prioritise for further operational testing?

The Week 5 work focused on data preparation, exploratory analysis, KPI calculation, visualisation, interpretation and business recommendations.

Data and Analytical Approach

The analysis used the HealthConnect appointment dataset and the accompanying data dictionary. The dataset contains 5,000 appointment records, 18 variables, and 1,696 unique patient IDs.



Analytical population

For the main no-show KPI, cancellations were excluded from the denominator. Eligible appointment opportunities were defined as Attended + No-Show. This produced 4,737 eligible records: 2,314 attended appointments and 2,423 no-shows. The 263 cancelled appointments were retained for separate operational analysis.

Data preparation and quality checks

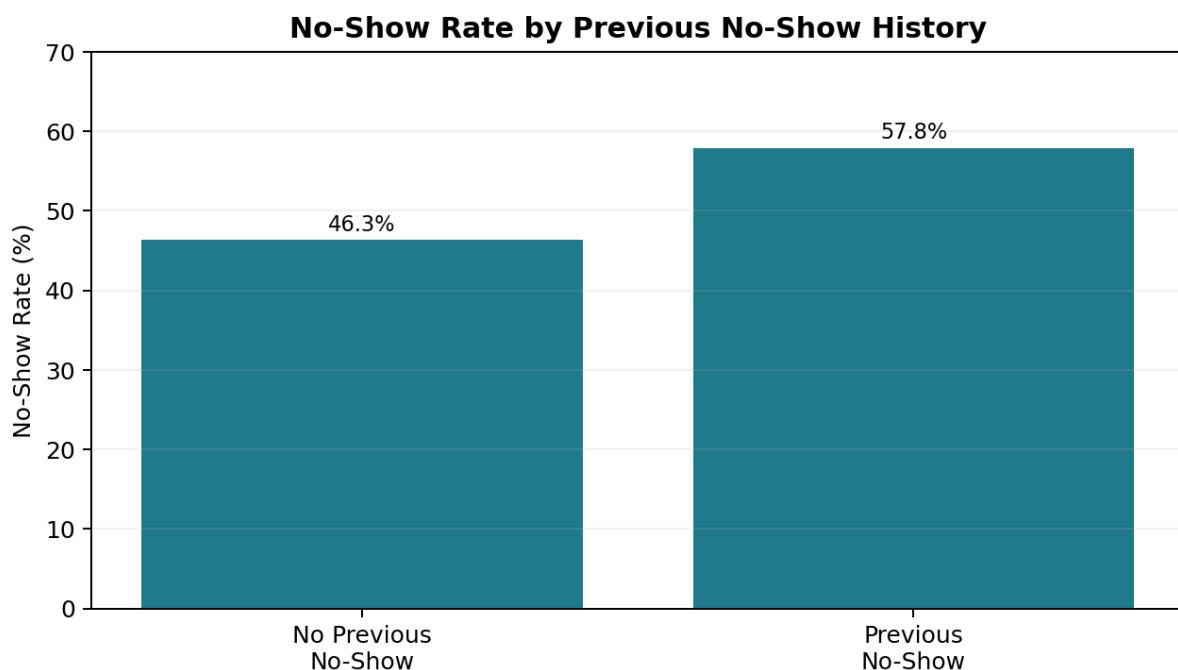
- No full-row duplicates or duplicate appointment IDs were identified; all 5,000 appointment IDs were unique.
- Booking and appointment dates were converted to datetime and no appointment date occurred before its booking date.
- The supplied booking_lead_days values matched the difference between booking date and appointment date for all records.
- Historical fields were logically consistent: previous_no_shows did not exceed previous_appointments.
- Age values and age-group labels were consistent across all six age bands.
- The original dataset contained 1,366 blank reminder_channel values, all corresponding to appointments where reminder_sent = No. These were treated as structurally expected and relabelled as No Reminder in the cleaned analytical dataset.
- Distance to clinic had 90 missing records (1.8%) and waiting time had 60 missing records (1.2%). These records were retained in the master dataset and excluded only from analyses requiring the relevant field.

KPI Results

KPI	Result	Business Question	Interpretation
Overall No-Show Rate	51.2%	What proportion of eligible appointments result in a no-show?	Establishes the baseline size of the attendance problem.
No-Show Rate by Previous History	57.8% vs 46.3%	Does previous no-show history relate to current attendance?	Previous no-show history is associated with an 11.5 percentage-point higher observed no-show rate.
No-Show Rate by Reminder Status	54.6% without vs 49.9% with reminder	Are reminded appointments associated with different attendance outcomes?	Reminded appointments show a 4.8 percentage-point lower observed no-show rate.
No-Show Rate by Booking Lead Time	29.5% (0-7 days) to 63.9% (31-60 days)	Does the time between booking and appointment relate to attendance?	No-show rate rises progressively as booking lead time increases.
No-Show Rate by Distance	48.7% (<5 km) to 60.5% (20+ km)	Does travel distance relate to attendance?	The longest-distance group records the highest observed no-show rate.

Exploratory Findings

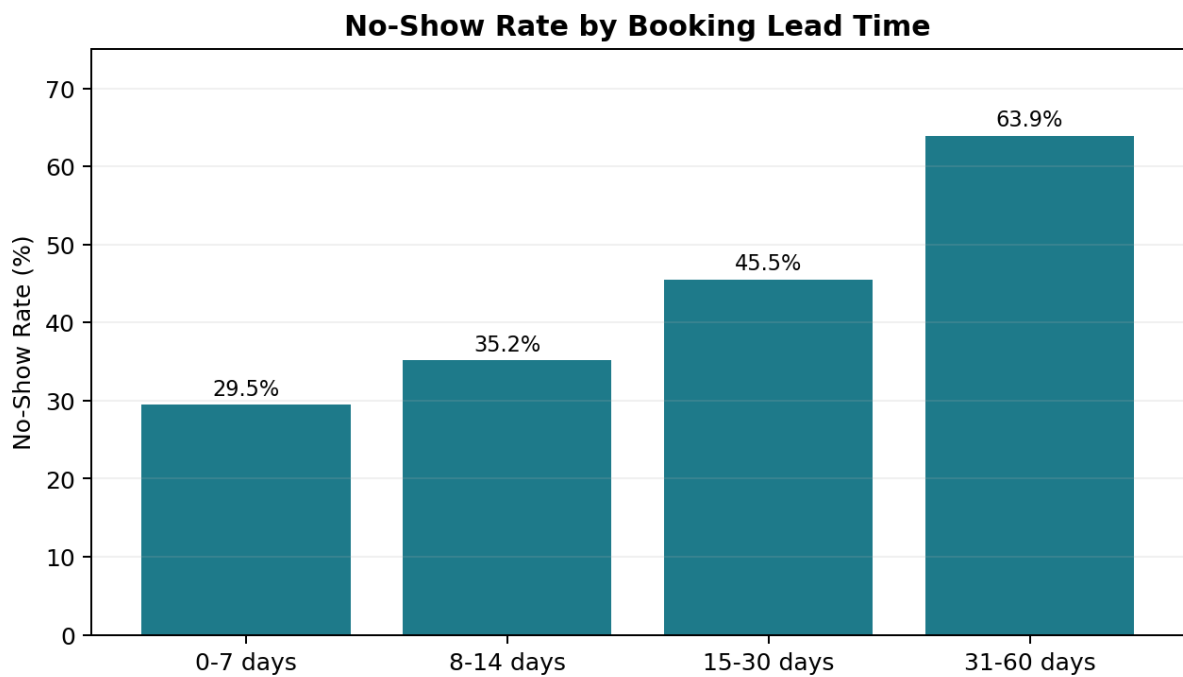
Previous no-show history



Appointments linked to patients with previous no-shows recorded a 57.8% no-show rate, compared with 46.3% among patients without previous no-shows. The difference of approximately 11.5 percentage points makes appointment history one of the clearest segmentation signals in the Week 5 analysis.

Business implication: Previous attendance behaviour could be used to prioritise confirmation or outreach, subject to appropriate safeguards and further testing.

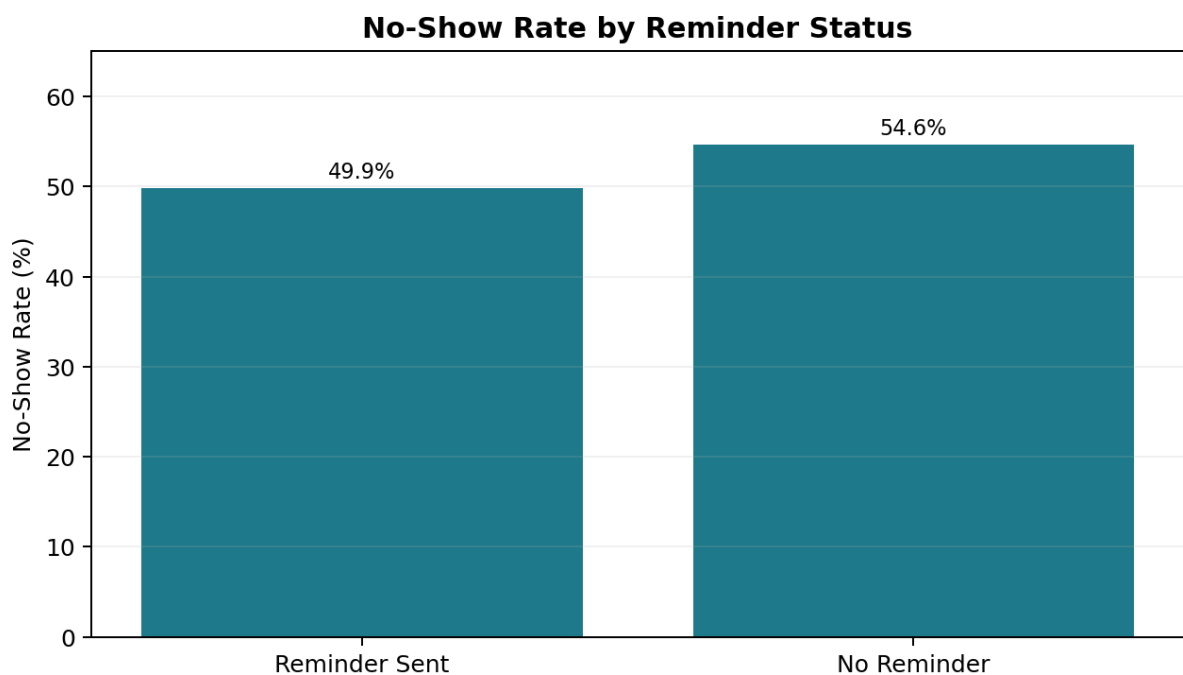
Booking lead time



The no-show rate increased from 29.5% for appointments booked 0-7 days in advance to 63.9% for appointments booked 31-60 days in advance. This was the strongest ordered pattern in the analysis. No eligible records were in the 60+ day group.

Business implication: Long-lead bookings are a strong candidate for reconfirmation closer to the appointment date.

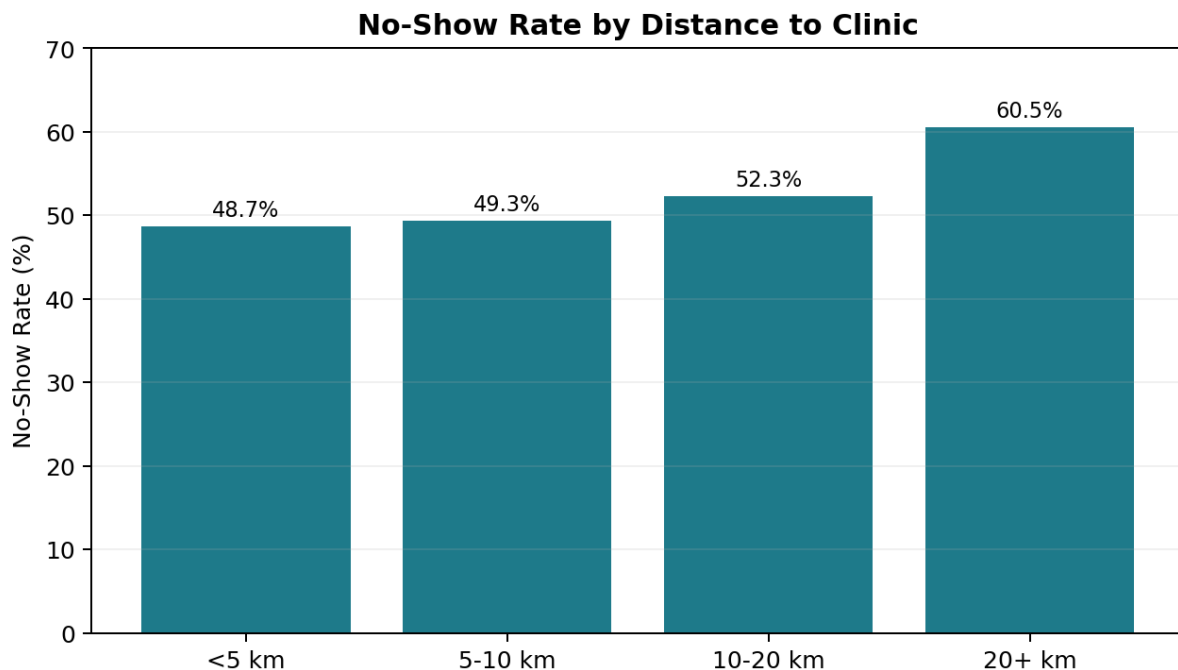
Reminder status and channel



Appointments without reminders recorded a 54.6% no-show rate, compared with 49.9% for appointments where a reminder was sent. By channel, SMS recorded the lowest observed no-show rate (48.0%), followed by email (51.0%) and WhatsApp (52.7%); appointments with no reminder had the highest no-show rate (54.6%).

Business implication: reminder activity appears useful, but the notebook correctly treats this as an association rather than proof that reminders cause better attendance. Reminder assignment may not be random.

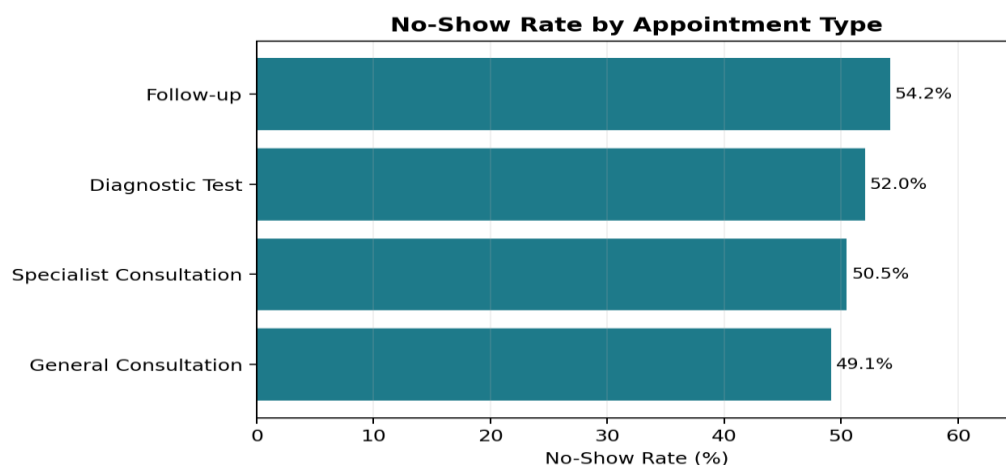
Distance to clinic



No-show rates rose across the distance bands, from 48.7% for patients travelling less than 5 km to 60.5% for patients travelling 20 km or more. The 20+ km group contained 375 eligible appointments.

Business implication: travel burden may be an important operational consideration when prioritising patient support, although the dataset does not capture all transport or socioeconomic factors.

Appointment type



Follow-up appointments had the highest observed no-show rate at 54.2%, followed by Diagnostic Tests at 52.0%, Specialist Consultations at 50.5% and General Consultations at 49.1%. Follow-up appointments also had substantial volume, making this segment operationally relevant.

Other Findings

Area	Observed Result	Interpretation
Appointment day	Monday was highest at 53.1%; Friday was lowest at 48.7%.	Differences exist but are smaller than the lead-time and history patterns.
Appointment time	Evening was highest at 52.6%; Afternoon 51.2%; Morning 50.7%.	Time-of-day differences are modest; volume should be considered alongside rate.
Age group	Rates ranged from 48.1% for 65+ to 53.0% for 55-64.	Differences are moderate and should remain descriptive, not discriminatory.
Waiting time	Most groups were close to 49.6%-52.2%; 60+ minutes was 66.7% but based on only 3 records.	The 60+ minute result is unstable; waiting time is less informative in this initial analysis.
Cancellations	Overall cancellation rate was 5.26%; Specialist Consultation was highest at 6.0%.	Cancellations are operationally relevant but are not equivalent to no-shows.

Consolidated Business Insights

- The overall no-show baseline is high: 51.2% of eligible attended/no-show appointment opportunities were missed.
- Previous no-show history is a strong segmentation signal: the observed no-show rate was 57.8% versus 46.3% for patients without prior no-shows.
- Booking lead time shows the clearest ordered pattern: no-show rates rose from 29.5% at 0-7 days to 63.9% at 31-60 days.
- Longer travel distance is associated with weaker attendance: patients 20+ km from the clinic recorded a 60.5% no-show rate.
- Reminders are associated with better attendance: appointments with a reminder recorded a 49.9% no-show rate compared with 54.6% without a reminder.
- Follow-up appointments had the highest service-category no-show rate at 54.2% and enough volume to merit workflow review.
- Appointment day, time, age group, and waiting time add context, but their differences are smaller or less stable than the strongest patterns above.

Business Recommendations

- Prioritise confirmation for patients with previous no-show history. Test additional confirmation steps or earlier outreach for appointments linked to previous no-shows.
- Introduce reconfirmation for long-lead appointments. Send a reconfirmation message closer to the appointment date for bookings made several weeks in advance.
- Review and test the reminder strategy. Assess reminder timing and channel effectiveness through a controlled pilot rather than assuming the observed association is causal.

- Consider access challenges for patients travelling longer distances. Where operationally feasible, test clearer travel information, confirmation outreach, or other administrative support for long-distance patients.
- Review follow-up appointment workflows. Investigate whether follow-up scheduling, communication, or timing differs from other appointment types.
- Prioritise using both rate and volume. A high percentage in a small group may be less operationally important **than a moderately high rate affecting many appointments.**

Limitations and Risks

- The dataset is fictional and anonymised, so findings should not be generalised directly to a real clinic.
- The analysis is observational; association does not establish causation.
- Distance and waiting-time analyses exclude a small number of records with missing values in those fields.
- Reminder assignment may not be random, so reminder-group differences may reflect other patient or appointment characteristics.
- The dataset does not include all real-world attendance drivers such as transport reliability, work schedules, socioeconomic constraints or acute health changes.
- Small groups should not be over-interpreted; the 60+ minute waiting-time group is a clear example.
- Demographic patterns should remain descriptive and should not be used to make discriminatory assumptions about individual patients.
- Waiting time should be interpreted cautiously if its timing relative to the attendance decision is not clearly established.

Week 6 Focus

- Refine the strongest analyses and check whether key patterns persist across subgroups.
- Review interactions such as booking lead time by reminder status or appointment type.
- Refine the Power BI dashboard and strengthen the business narrative.
- Validate KPI definitions and calculation logic.
- Incorporate genuine cross-track feedback and document the exchange.
- Test whether proposed recommendations remain supported across relevant segments.
- Prepare a clearer final analytical story for subsequent project stages.

Conclusion

Week 5 moved the HealthConnect Data Analytics track from problem definition into practical implementation. The data was prepared and validated, cancellation handling was defined, five core KPIs were calculated, and exploratory analysis was completed across patient history, reminders, booking lead time, distance, appointment timing, appointment type, age group and waiting time. The clearest observed patterns are associated with previous no-show history, longer booking lead times, and greater distance to the clinic, with reminder status also showing a potentially useful difference. These findings provide evidence-based areas for further testing and dashboard development while remaining descriptive associations within the supplied fictional dataset.